Annie Prasanna Manoharan

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https://github.com/Annie0159

Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: Graph algorithms, Protocol Buffers, WebSockets, Microservices, REST APIs, Grafana, Docker, Linux, CI/CD, OpenStreetMap
Data Engineering: Apache Kafka, Apache Flink, PostgreSQL
Languages: SQL, C++, Python, C
Machine Learning and AI: Flask

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation

Projects

Road Network Route Planning Engine *C++, Python, OpenStreetMap*

- Implemented a road network graph in C++ from OpenStreetMap map extracts, modeling intersections as nodes and road segments as weighted edges
- Built Dijkstra and A* shortest-path search over the graph with heuristics based on geographic distance, and compared runtime behavior across route queries of varying path length
- Added a Python benchmarking harness to profile query latency and memory use across different graph sizes

Pacman Search Algorithms (coursework) *Python, BFS, A*, admissible heuristics*

- Implemented custom problem states, successor functions, goal conditions, and state representations for maze-based environments
- Implemented BFS and A* with admissible heuristics, and a greedy approximation strategy for large state spaces where optimal search was too expensive